

ETHAN C. GOLDSTEIN

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Computer Information Systems student working across systems programming, applied machine learning, and signal processing. Active Public Trust clearance with federal records experience. Validates every result by trying to break it first.

EDUCATION

University of South Carolina

B.S. Computer Information Systems, Minor in Business Information Management

Coursework: Data Structures and Algorithms (Java), Information Security Principles, Web Applications, Capstone Project

Study Abroad: Florence University of the Arts, Firenze, Italy

Columbia, SC

Expected May 2027

Spring 2026

EXPERIENCE

AI & Workflow Automation Engineer

2025 to Present

Independent Practice · github.com/ethan-goldstein/Autonomous-OS

Remote

- Designed an 11-agent orchestration platform on a Node and Express ESM core where every agent inherits scheduling, SQLite persistence, and live log streaming from a shared base class, so adding an agent is one file and one registry line.
- Built a two-tier LLM dispatcher, headless Claude Code CLI primary with local Ollama fallback behind a 10-minute persisted circuit breaker, and capped inference at 2 concurrent processes on a FIFO semaphore after unbounded forking crashed an 8GB machine.
- Persisted the fleet on node:sqlite with no ORM across 13 tables, adding WAL journaling and a 5-second busy timeout after concurrent writes began failing boots, and streamed live state to a React dashboard over Server-Sent Events.
- Routed every outward-facing action through one egress queue where risky lanes require a human tap that no toggle can override, and funnelled all model calls through a single chokepoint that treats fetched content as data and never as instructions.
- Wrote HMAC-SHA256 session auth on node:crypto alone behind a prefix allowlist mounted before any route is declared, with no localhost bypass because Tailscale terminates TLS on loopback; deployed via launchd with no public port.
- Made automated outreach defensible: per-lane daily caps dripped across staggered windows, randomized send jitter, DNS MX pre-validation that fails open, and an IMAP bounce loop retiring dead addresses.

GovCIO

2026 to Present

Data Processor

Tysons Corner, VA

- Hold an active Public Trust clearance (U.S. Treasury, IRS) extended to Department of Veterans Affairs systems by reciprocity.
- Process confidential Veteran records under the VA HTMS contract in secured remote environments.

Oxford Government Consulting

2025

Data Entry Operator

Tysons Corner, VA

- Supported the IRS Digitalization-as-a-Service program with GovCIO, on a pipeline processing 2 million document images per day.

The Baseball Zone, Manager. Rebuilt the company website; ran front desk operations, scheduling, and daily finances. 2021 to 2024

PROJECTS

PARALLAX: Bitemporal Analytical Engine C++20, WebAssembly, SIMD, libFuzzer

- Query language written end to end, lexer through cost-based planner, over a columnar bitemporal store with a Z-order index. Policy refusal at plan time, sound because two of four cardinality estimators are exact. Differential tested against a naive oracle.

NULLHYP: Quantitative Research Engine TypeScript, Web Workers, zero-dependency ML, Python cross-check

- Reports no edge: Sharpe 2.46 in sample, -0.18 out, deflated Sharpe 0.06 over 12 trials. Validation came before any model: purged K-fold with embargo, combinatorial purged cross-validation, and a lookahead audit that recomputes every feature on data truncated at bar t. Trees, elastic net, and logistic regression from scratch in a Web Worker, matched on 22 of 22 checks by a pandas build.

WiFi Sensing Lab Python, signal processing, statistics, privacy engineering

- The 0.5 to 3 Hz gait band is unobservable at consumer polling rates, so motion detection from RSSI alone moved to the 0.1 to 0.8 Hz body-shadowing envelope, each link learning its own noise floor from a tracker that falls fast and rises slowly. Median AUC 0.676 over 10 links against a permutation null at 0.503, $p = 0.017$. The public repository contains no network client.

Real-Time Multiplayer Engine WebSockets, Cloudflare Durable Objects, Three.js, MediaPipe

- Authoritative game server on edge compute: fixed 60Hz simulation with seeded RNG decoupled from render. Three public titles.

Vector Stroke-Rendering Engine React, Vite, Three.js, GitHub Actions

- Auto-traces a raster logo into vector contours with marching squares, replayed against a Three.js pen. Live on a paying client's site.

SKILLS

Languages & Verification: C++20, Java, Python, TypeScript, SQL, Bash; libFuzzer, ASan and UBSan, differential testing

AI, ML & Statistics: multi-agent orchestration, local LLM inference (Ollama), Claude Code CLI, prompt-injection mitigation; gradient boosting, elastic net and logistic regression from scratch, purged and combinatorial cross-validation, permutation tests, DSP

Backend & Distributed Systems: Node.js, Express, WebSockets, Server-Sent Events, node:sqlite (WAL), REST, OAuth 2.0, HMAC-SHA256 auth, Cloudflare Workers and Durable Objects, Tailscale, launchd and cron, Playwright, Git, GitHub Actions CI/CD

Frontend: React, Next.js App Router, Vite, Three.js, WebGL, WebAssembly, GSAP, MediaPipe, code splitting, WCAG accessibility

Clearance: Public Trust (Active), U.S. Treasury (IRS); VA reciprocity 2026, Veteran and Enterprise Technology Sector